Carleton University

Department of Systems and Computer Engineering

SYSC 5108 ***Deep Learning*** ***Winter 2026***

Course Outline

Instructor:
Professor C. Huang
Room 4486ME
Tel: 520-2600 ext. 5730
Email: changchenghuang@cunet.carleton.ca
Website: http://www.sce.carleton.ca/faculty/huang.html

Office Hours
TBD

Calendar Description
The course covers a variety of deep learning techniques with rigorous analysis and practical implementations. Deep supervised and unsupervised learning. Deep reinforcement learning. Directed and undirected probabilistic models. Representation learning. Monte Carlo methods and deep generative models. Implementations and case studies.

Prerequisites
Undergraduate calculus, linear algebra, probability and statistics; Programing experience in Python is desired.

Course Objectives
The goal of the course is to provide a comprehensive coverage on modern machine learning theories and techniques. Emphasis is placed on the understanding of the analytical backgrounds of various models and hands-on experience of implementing and running machine learning algorithms. The course includes a brief overview of probability theory.

Texts: I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning, MIT Press, 2016 and S. Raschka and V. Mirjalili, *Python Machine Learning*, Packt Publishing, 3rd Edition, 2019

References:

T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Springer, Second Edition, 2008

S. Marsland, *Machine Learning: An Algorithm Perspective*, Chapman and Hall/CRC, 2014

L. Graesser and W. L. Keng, *Foundations of Deep Reinforcement Learning: Theory and Practice in Python*, Pearson Education 2020

Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction,* MIT Press, 2nd Edition 2018

H. Perros, *Computer Simulation Techniques: The definitive introduction!*
https://people.engr.ncsu.edu/hp/files/simulation.pdf

Marking Scheme

Engagement	15%

Assignments	10%
Peer Review	10%
Project	25%
Final Exam	40%

Breakdown of Course Requirements

a. Project requirements
   1. Projects can be done in teams.
   2. Each team can have at most two students.
   3. Each project will include two deliverables: one in presentation, one in final report.
   4. Students can select any application area for project.
   5. Project must use at least one deep learning algorithm.
   6. Project must present training and test performance results with adopted deep learning algorithm.

b. Attendance

   Students are expected to attend all lectures. The University requires students to have a conflict-free timetable. For more information, see the current Graduate Calendar, Academic Regulations of the University, Section 2.1.3, Course Selection and Registration and Section 2.1.7, Deregistration.

c. Deferred Term Work

   Students who claim illness, injury or other extraordinary circumstances beyond their control as a reason for missed term work are held responsible for immediately informing the instructor concerned by submitting the academic consideration for coursework form (https://carleton.ca/registrar/academic-consideration-coursework-form/) and for making alternate arrangements with the instructor and in all cases this must occur no later than three (3) days after the term work was due. The alternate arrangement must be made before the last day of classes in the term as published in the academic schedule. Consult https://calendar.carleton.ca/grad/gradregulations/.

d. Final Exam

   - Final exam will take the form of in-person assessment.
   - Final exam is for evaluation purpose and will not be returned to students.
   - Final exam will be a closed-book exam. Only non-programmable calculator is allowed.

e. Students who are unable to write the final examination because of a serious illness/emergency or other circumstances beyond their control may apply for accommodation by contacting the Registrar's office. Consult the https://calendar.carleton.ca/grad/gradregulations/#examinations

f. Deferred Final Exam:

   Students who encounter circumstances beyond their control may defer their final examination to write it at a later date. Deferral applications for final examination must be made using the Online Deferral Application (https://carleton.ca/registrar/deferral/) no more than three business days after the original final examination. Supporting documentation must be provided.

g. Deadline Policy

All assignments shall be submitted online. Late assignments will not be accepted.

h. Checking Marks

Term marks will be posted as soon as they are available. It is each student's responsibility to check that marks are correct or report any errors by the last class.

Class Schedule
11:35am-12:55pm, Tuesday and Thursday, online

Course Schedule:

Week 1: Introduction to machine learning and deep learning. Definition, taxonomy, and history. Linear regression, basis expansion. K nearest neighbor. Regularization and capacity.

Week 2: Statistical machine learning basics. Maximum likelihood estimator. Logistic regression model. Stochastic optimization.

Week 3: Introduction to deep learning. Deep feedforward networks. Regularization methods. Adaptive learning rates and initialization strategies.

Week 4: Special neural networks, CNN, RNN, LSTM, Transformer. Practical methodology, hyper-parameter selection, distributed implementation.

Week 5: Reinforcement learning. Markov Chain. Dynamic programming.

Week 6: Q-learning. Policy Gradient Theorem. Actor-critic model.

Week 7: Unsupervised learning basics, K-means, PCA. Autoencoders, denoising autoencoder (DAE).

Week 8: EM algorithm, Structured probabilistic models, directed and undirected models. Monte Carlo methods, MCMC.

Week 9: Partition function, contrastive divergence (CD), stochastic maximum likelihood (SML).

Week 10: Deep generative network, RBM, DBN, DBM, generative adversarial networks (GAN).

Week 11: Student project presentations

Week 12/13: Student project presentations. Review.

General Regulations

Student Responsibility: It is the student's responsibility to remain informed of all rules, regulations and procedures required by their program and by the Faculty of Graduate and

Postdoctoral Affairs. Ignorance of regulations will not be accepted as a justification for waiving such regulations and procedures.

Academic Integrity: Students should be aware of their obligations with regards to academic integrity. Please review the information about academic integrity at: https://carleton.ca/registrar/academic-integrity/. This site also contains a link to the complete Academic Integrity Policy that was approved by the University's Senate.

Plagiarism:

- Please consult the Faculty of Engineering and Design information page about the Academic Integrity policy and our procedures (https://carleton.ca/engineering-design/current-students/fed-academic-integrity). Violations of the Academic Integrity Policy will result in the assignment of a penalty such as reduced grades, the assignment of an F in a course, a suspension or, expulsion.
- One of the main objectives of the Academic Integrity Policy is to ensure that the work you submit is your own. As a result, it is important to write your own solutions when studying and preparing with other students and to avoid plagiarism in your submissions. The University Academic Integrity Policy defines plagiarism as "presenting, whether intentionally or not, the ideas, expression of ideas or work of others as one's own." This includes reproducing or paraphrasing portions of someone else's published or unpublished material, regardless of the source, and presenting these as one's own without proper citation or reference to the original source.
- Examples of violations of the policy include, but are not limited to:
    - any submission prepared in whole or in part, by someone else;
    - using another's data or research findings without appropriate acknowledgement;
    - submitting a computer program developed in whole or in part by someone else, with or without modifications, as one's own;
    - failing to acknowledge sources of information through the use of proper citations when using another's work and/or failing to use quotation marks; and
    - unless explicitly permitted by the instructor in a specific course, the use of generative AI and similar tools to produce assessed content (such as text, code, equations, images, summaries, videos, etc.).
- AI tools can be used to help your learning such as understanding concepts, algorithms, architectures, design principles, etc. However, violations of academic integrity standards include copying and/or paraphrasing the content generated from AI tools without permission by instructors and submitting them as your own original work for grade items. As our understanding of the uses of AI and its relationship to student work and academic integrity continue to evolve, students are required to discuss their use of AI in any circumstance not described here with the course instructor to ensure it supports the learning goals for the course.

Copyright on Course Materials: The materials created for this course (including the course outline and any slides, posted notes, labs, project, assignments, quizzes, exams and solutions) are intended for personal use and may not be reproduced or redistributed or posted on any web site without prior written permission from the author(s).

Health and Safety: Every student should have a copy of our Health and Safety Manual. A PDF copy of this manual is available online: http://sce.carleton.ca/courses/health-and-safety.pdf

Students from the University of Ottawa: You can request to have access to cuLearn: please see http://gradstudents.carleton.ca/forms-policies/